

QUICK ACTIVITY 1: SCHOOL SUPPLIES CALCULATOR

PROBLEM ANALYSIS: A student needs to calculate the cost of buying several notebooks and pens. The cost depends on the number of notebooks, the price per notebook, the number of pens, and the price per pen.

DECOMPOSITION: For this problem, identify the inputs, calculate the notebook cost, calculate the pen cost, calculate the total cost, calculate the change, and display the result.

Input	process	Output
number of notebooks number of pens payment	notebook cost = notebooks $\times$ 45 pen cost = pens $\times$ 12 total cost = notebook cost + pen cost change = payment - total cost	Total cost change

- ① Read the notebooks, pens, and payment.
- ② Compute the notebook cost.
- ③ Compute the pen cost.
- ④ Add them for the total cost.
- ⑤ Compute the change.
- ⑥ Display the total cost and change.

PSEUDOCODE

START

```

READ notebooks, pens, payment
notebookCost ← notebooks  $\times$  45
penCost ← pens  $\times$  12
totalCost ← notebookCost + penCost
change ← payment - totalCost
DISPLAY totalCost, change
    
```

END

DRY RUNS

Test Case 1:

3 notebooks, 2 pens, ₱200

STEP	VARIABLE	CALCULATION	RESULT
1	notebookCost	$3 \times ₱45$	₱135.00
2	penCost	$2 \times ₱12$	₱24.00
3	totalCost	₱135 + ₱24	₱159.00
4	change	₱200 - ₱159	₱41.00

TOTAL COST: ₱159.00
change: ₱41.00

Test Case 2:

2 notebooks, 3 pens, ₱150

STEP	VARIABLE	CALCULATION	RESULT
1	notebookCost	$2 \times ₱45$	₱90.00
2	penCost	$3 \times ₱12$	₱36.00
3	totalCost	₱90 + ₱36	₱126.00
4	change	₱150 - ₱126	₱24.00

TOTAL COST: ₱126.00
change: ₱24.00